On Regulatory and Organizational Constraints in Visualization Design and Evaluation

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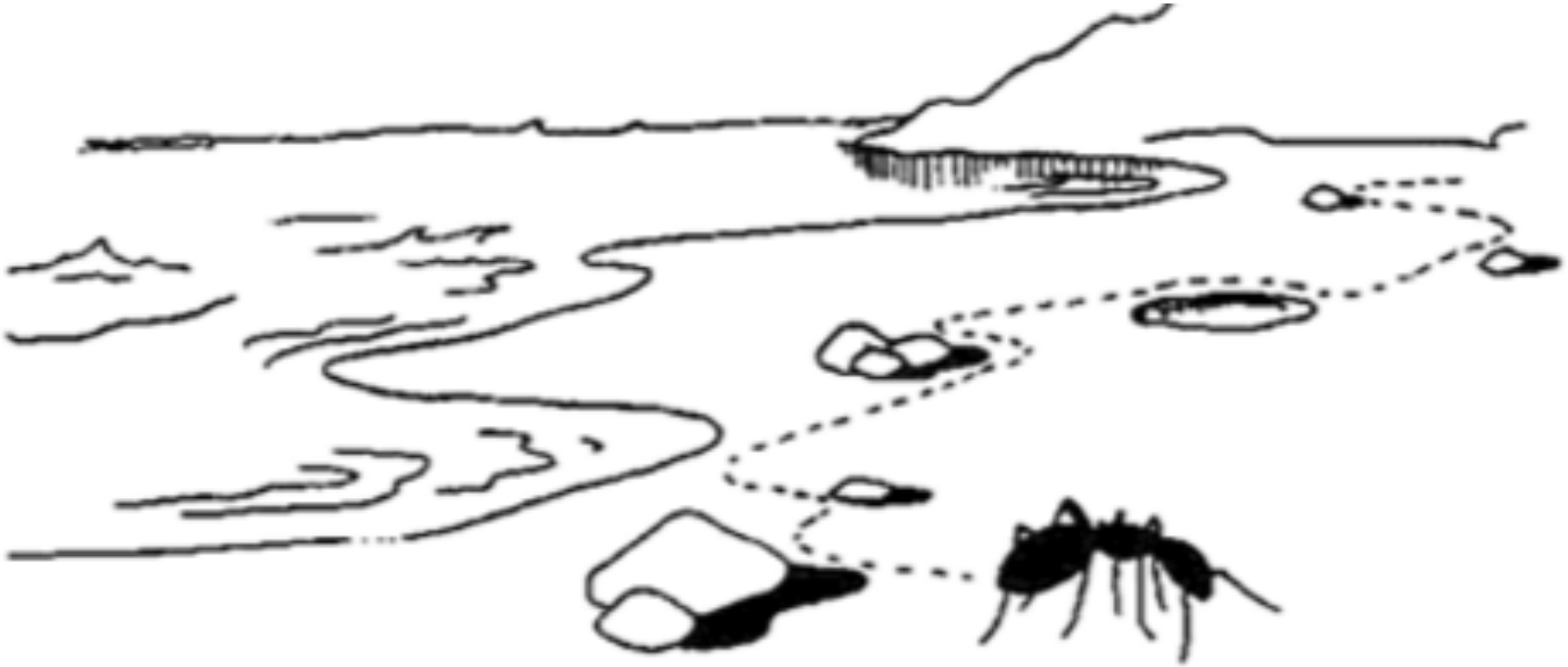
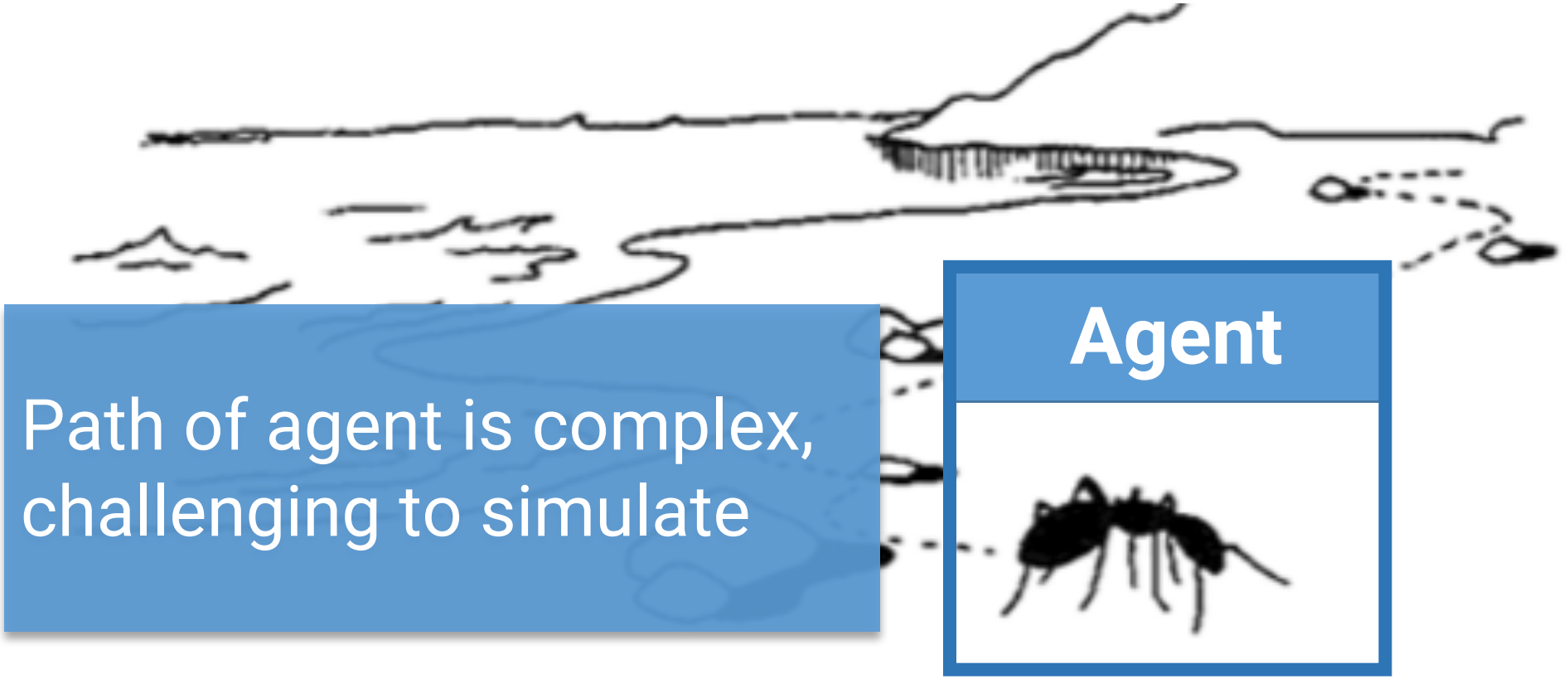
PARABLE OF THE ANT ON THE BEACH



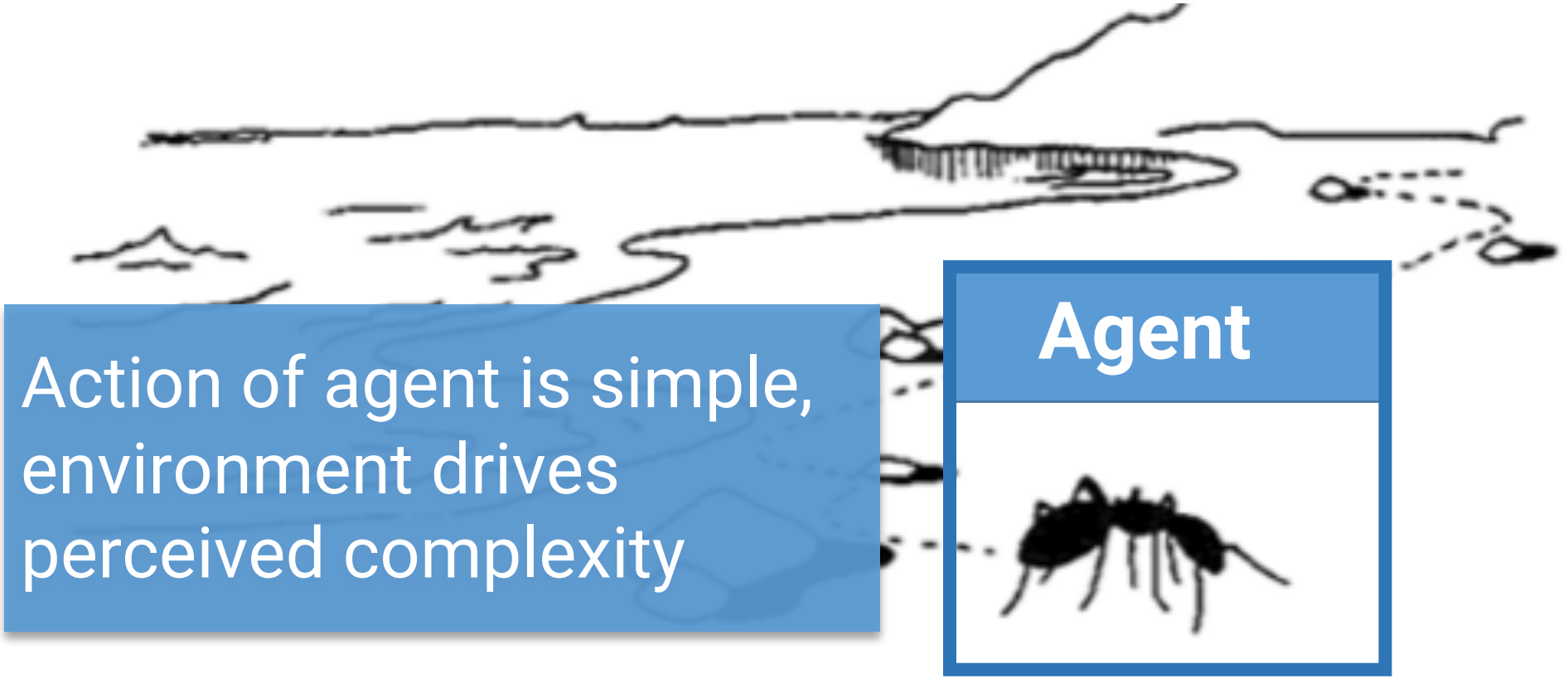
Simon. The Sciences of the Artificial, 1981

PARABLE OF THE ANT ON THE BEACH



PARABLE OF THE ANT ON THE BEACH

Environmental / Contextual Factors



Action of agent is simple, environment drives perceived complexity

VIS DESIGN & EVALUATION MISSING CONTEXT

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Please refer to paper for Case Study

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- **Contribution**: Explicit guidance to incorporate regulatory and organizational contextual constraints in Vis design and evaluation
- [How] Modifications to a Design Study Methodology
- [How] Demonstration with case study in healthcare
- **Value:** Prepare Vis research for highly regulated environments

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TYPES & IMPACTS OF EXTERNAL CONSTRAINTS

Regulatory : Data use and access as encoded in the Law

- Very important when working with individual-level person data

Organizational: Policies and practices enforced by an organization

- Includes *enforcement* of regulatory constraints
- Varies across organizations (some stricter, others not)

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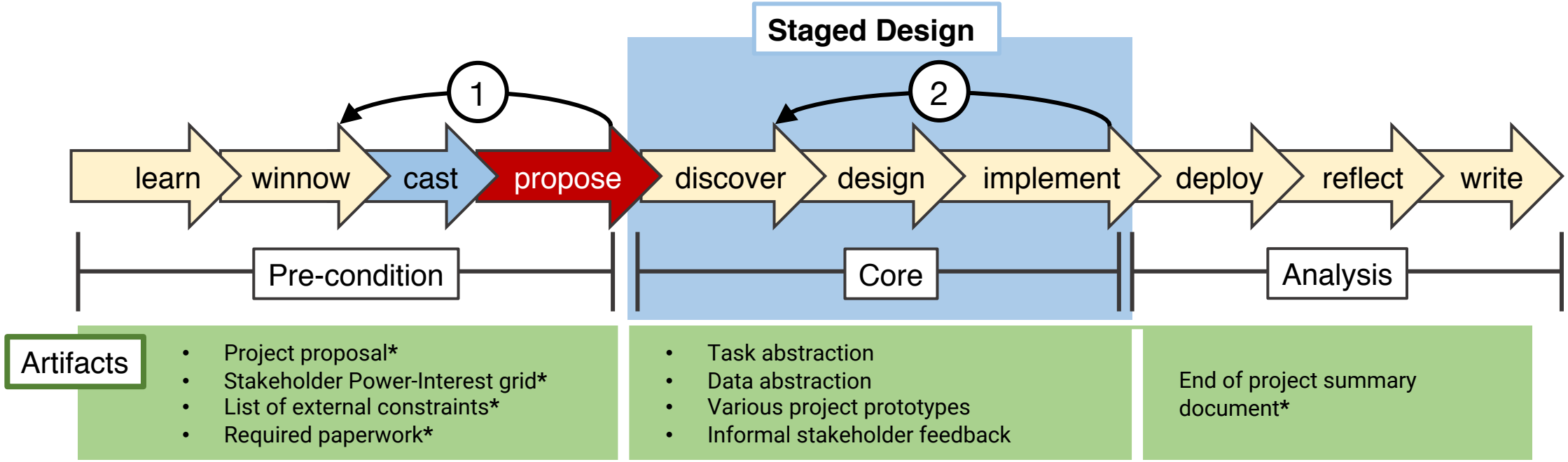
Example : Hypothesis *Generation* Considered Harmful

- Uncontrolled hypothesis generation can be illegal
  - Can analyze data for one purpose only at a time
  - Why: individual bears the burden of accidental disclosure
- Counter to Vis insight generation principle
- Vis methods must adapt to this constraints

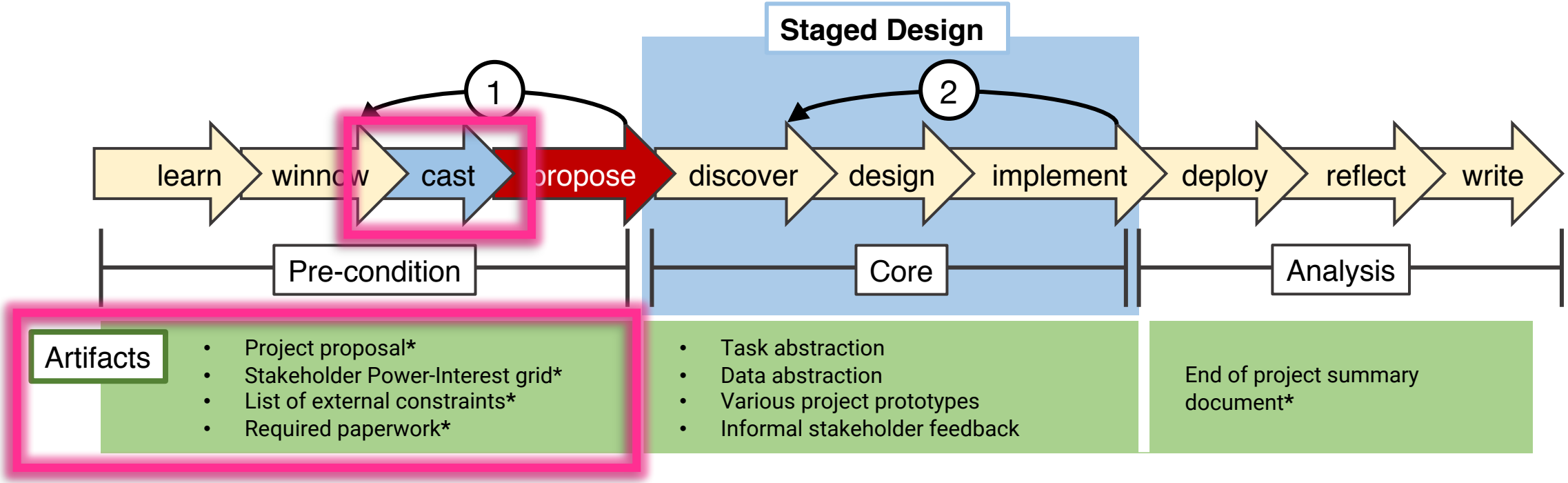
MODIFYING A DSM FOR HIGHLY REGULATED CONTEXT



Modify DSM by Sedlmair *et. al*. (2012) to include:

- Creation of additional artefacts
- Modified + added steps
- Mandatory (as opposed to suggested) check backs
- Planning the staging of Vis deliverables
- Using synthetic data

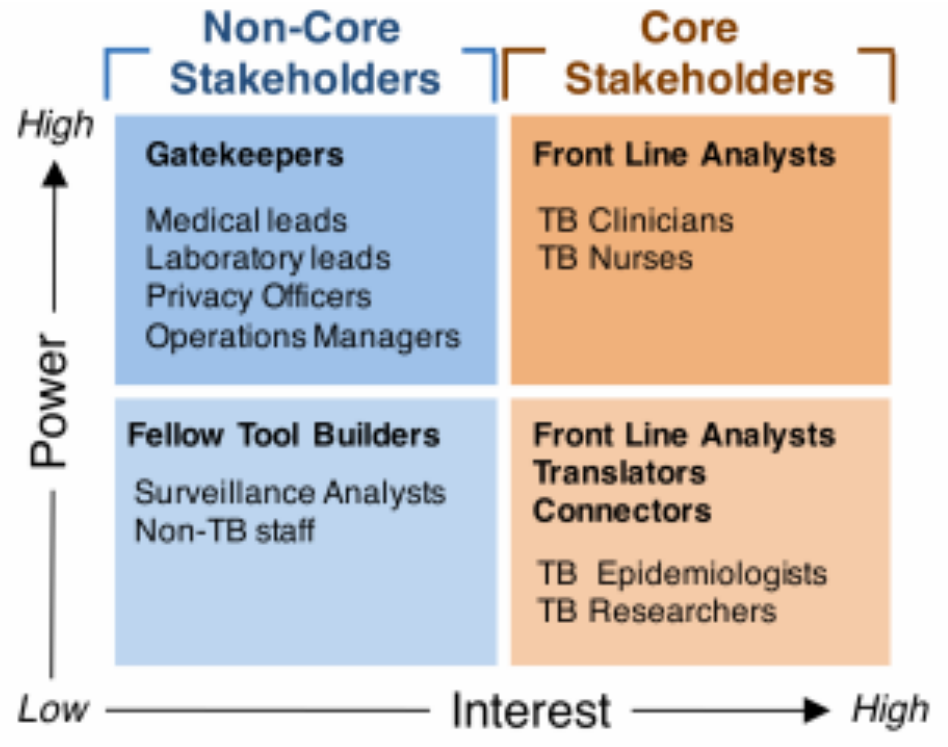
R1: CLASSIFY USERS BY POWER AND INTEREST



<u>Recommendation 1</u>

Classify stakeholders according to power over and interest in project outcomes

R1 EXAMPLE FROM HEALTHCARE CASE STUDY



- Core stakeholders may not be aware of all Gatekeepers
- Use grid to find “unknown” Gatekeepers

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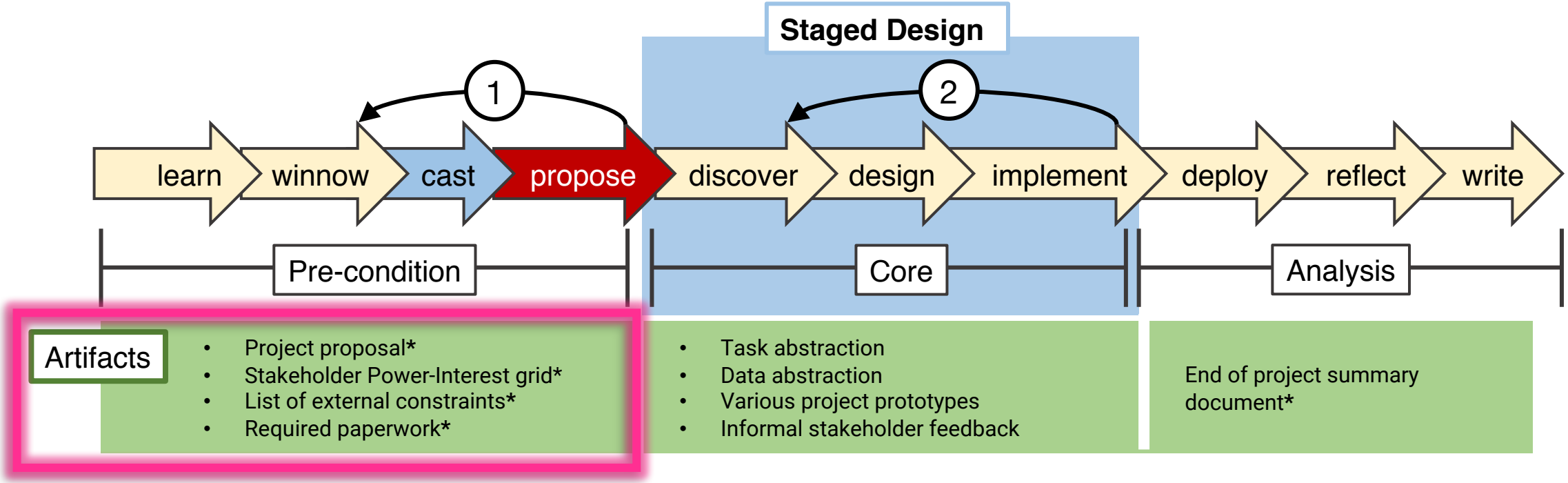
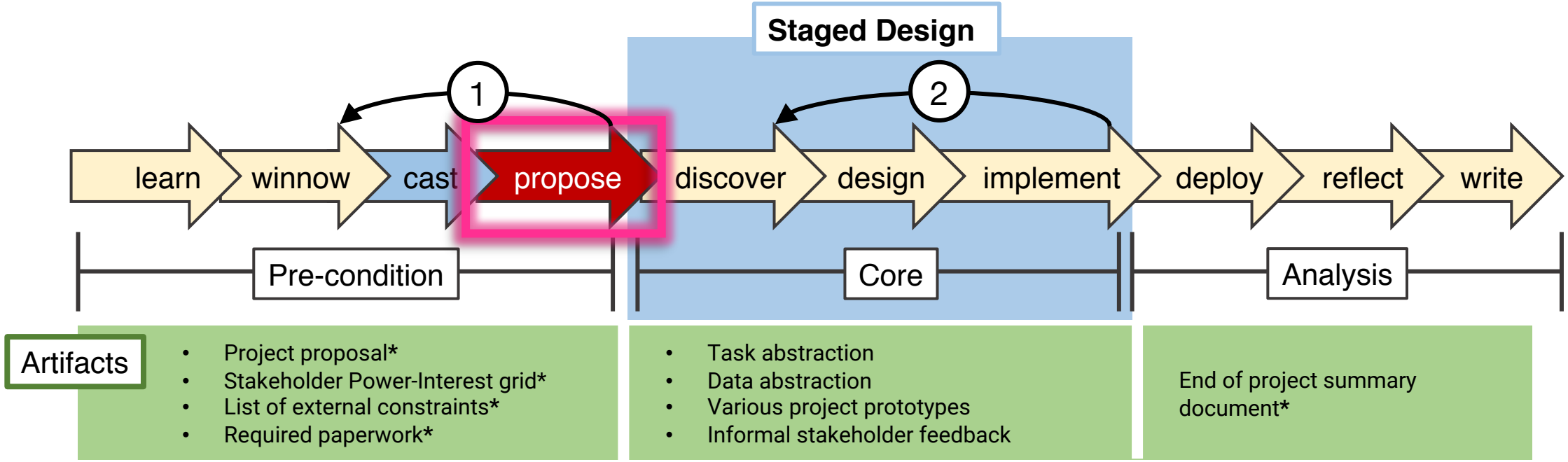
R2: ACTIVELY MANAGE COMMUNICATION



<u>Recommendation 2</u>

Actively manage communication with stakeholders using the power-interest grid

R3: PROPOSAL DOCUMENT

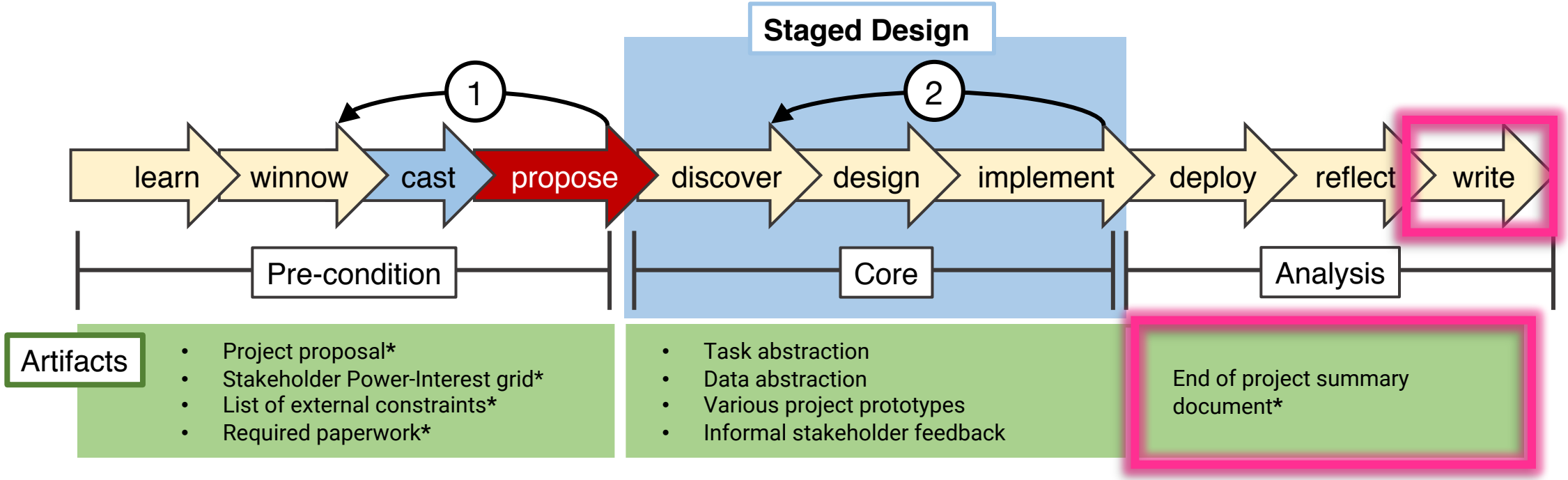


Recommendation 3

Create a formal proposal document

Lays out stakeholder & researcher expectations

R4: PROJECT SUMMARY DOCUMENTS

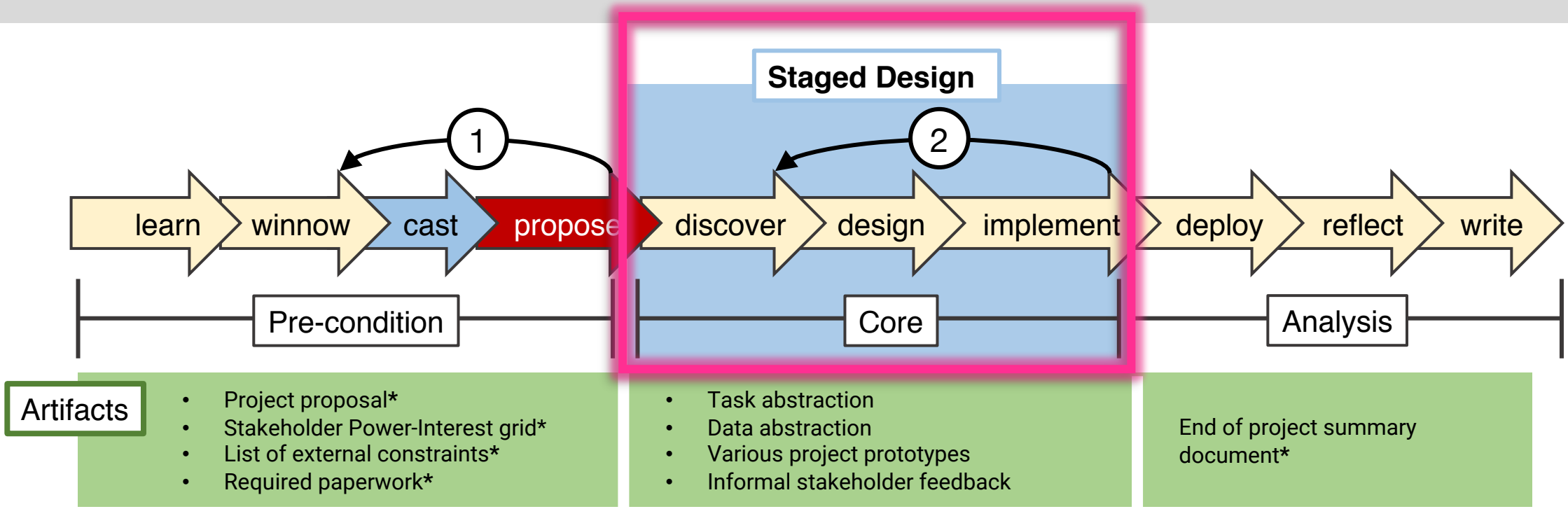


<u>Recommendation 4</u>

Create a summary document at the end of a project

Demonstrates value of VIS research

R5: STAGED DESIGN PROCESS

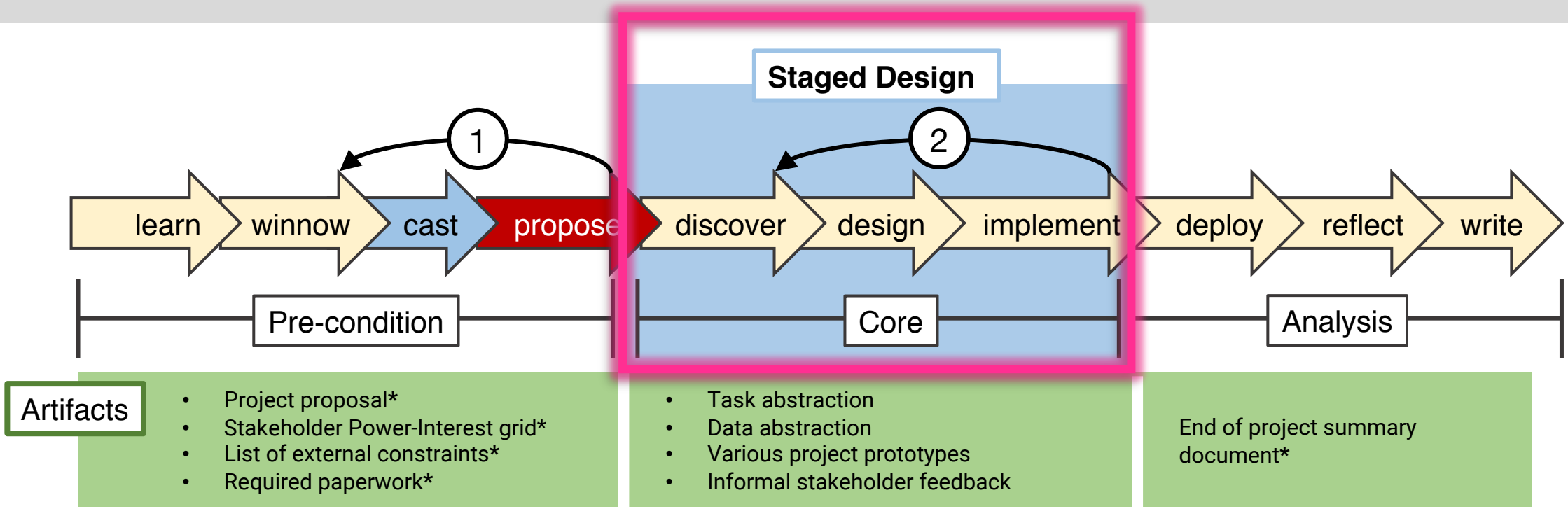


<u>Recommendation 5</u>

Plan staged design deliverables around emerging constraints

Re-evaluate project viability at end of each stage

R6: SYNTHETIC DATA



Recommendation 6

Use carefully constructed synthetic data

Get buy-in to use real data later

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Identify, classify, mitigate, and evaluate external constraints

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MITIGATE

Engender trust of Vis methodologies

R2: Activity manage communicate with stakeholders

R3: Create a formal proposal document

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EVALUATE

Model constraints in design and evaluation

R5: Use a staged development process

R6: Use synthetic data early on if real data is not immediately available

CONCLUSIONS

- Regulatory & Organizational constraints are not insurmountable
- Concrete recommendations as first steps
- Vis research in regulated contexts is rewarding & valuable – go forth and visualize, but be prepared!

Paper & Contact Info

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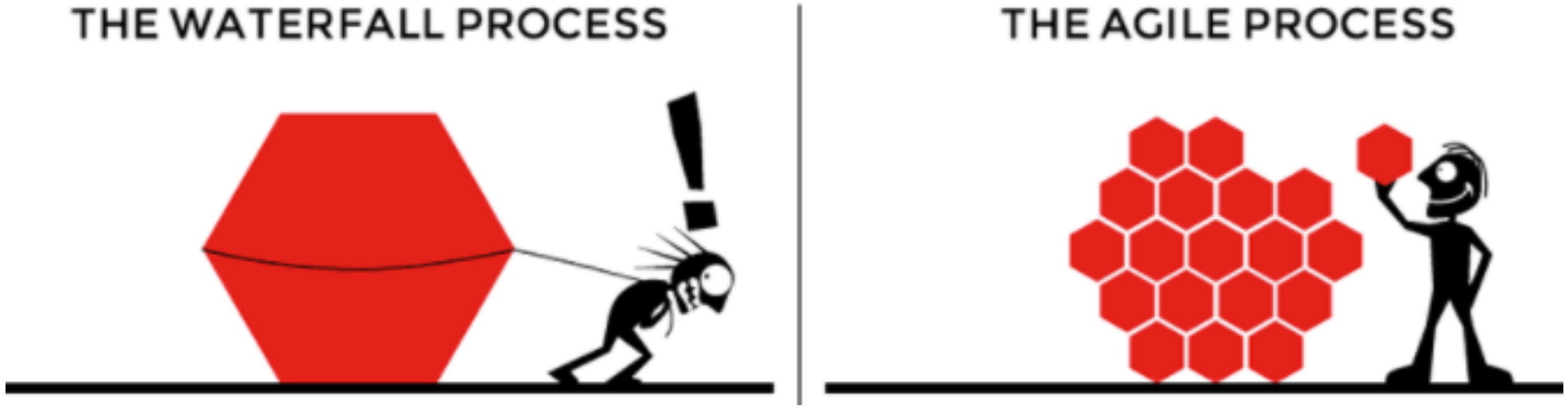
Additional Slides

TYPES & IMPACTS OF EXTERNAL CONSTRAINTS

Example – Restrictions due to Organization constraints

Agile Development Considered Harmful

- Design Study Methodologies are agile development methods
- Benefits of Agile methods overlooked in regulated contexts
- Why: Policy dictates having all requirements and deliverables before development work begins